MATH0010 Mathematical Methods 1

Year:	2026–2027
Code:	MATH0010
Level:	4 (UG)
Normal student group(s):	UG: Y1 Mathematics programmes Year 1 MEng Mathematical Computation
Value:	15 credits (7.5 ECTS credits)
Term:	1
Assessment:	70% examination, 15% coursework, 15% departmental test(s). The departmental tests are divided into 5% online Elementary Techniques Test and 10% midsessional exam in January. In order to pass the module you must have at least 90% on the Elementary Techniques Test, 40% for the examination mark, and 40% for the final weighted mark.
Normal Pre-requisites:	A* in A-level Mathematics and Further Mathematics
Lecturer:	Prof V Smyshlyaev

Course Description and Objectives

The aim of the course is to bring students from a background of diverse A-level syllabuses to a uniform level of confidence and competence in vectors, complex numbers, calculus and differential equations. The course covers vectors, complex numbers, standard functions of a real variable, methods of integration, ordinary differential equations and probability. Each topic is given a formal treatment and illustrated by examples of varying degrees of difficulty. The elementary techniques test will be run online and may be taken multiple times during the term. It will consist of 10 basic calculational questions; you must get at least 9 correct answers to pass. It is necessary to pass this test in order to pass the module.

Recommended Texts

1. Safier, *Precalculus*
2. Spiegel, *Advanced Calculus*
3. Ayres, *Differential Equations* (Schaum's Outlines)
4. Gilbert, Jordan and Towers, *Guide to Mathematical Methods* (Palgrave Macmillan)
5. Riley, Hobson, and Bence, *Mathematical Methods for Physics and Engineering* (Cambridge)
6. Kreyszig, *Advanced Engineering Mathematics* (Wiley)
7. Stephenson, *Mathematical Methods for Science Students* (Prentice Hall)
8. Stirzaker, *Probability and Random Variables: A Beginner's Guide* (Cambridge)

Detailed Syllabus

– Vector algebra: Scalar and vector products including triple products. Cartesian components. Applications to 3-dimensional geometry: lines and planes.

- Complex numbers: Argand diagram, loci, roots of unity, geometry.
- Revision of simple functions: Powers, exponentials, trig, hyperbolic. Differentiation. Maclaurin and Taylor series. Elementary properties of plane curves, curve sketching.
- Systematic revision of integration: Partial fractions, by parts, substitution. Definite and indefinite integrals. Improper integrals.
- Functions of several variables: Surface sketching, partial and directional derivatives, gradient of a function, level surfaces and tangent planes.
- Introduction to ordinary differential equations: First order (linear and non-linear). Second order reducible to first order. Linear equations of second and higher order (particular integral and complementary function).
- Probability: Sets. Sample spaces. Probability. Binomial distribution. The Poisson distribution. Normal distribution.

August 2026